

A
♠ **Sum Groups 10**

THE CHALLENGE: Break this whole square into groups of two or three numbers that add up to 10. A group's numbers must share sides. Is there only one way to solve this?

1	5	3	2
4	3	7	4
5	3	5	6
3	4	1	4

♠

A
♦ **Consecutive Numbers 2**

THE CHALLENGE: Place the numbers from 1 to 8 so the boxes for consecutive numbers do not share a side or touch diagonally.

1 2 3 4 5 6 7 8 ♦

A
♥ **Treasure Map 2**

THE CHALLENGE: When standing on a square, move the given number of squares, and only move to the right, left, up, or down. Find a route from the red Start square to the \$\$.

Start	1	2	1
2	3	2	3
3	1	2	3
1	3	1	\$\$

♥

A
♠ **Sum Groups 9**

THE CHALLENGE: Break this whole square into groups of two or three numbers that add up to 9. A group's numbers must share sides. Is there only one way to solve this?

5	4	3	6
7	4	2	3
2	5	3	6
8	1	1	3

♠

2
♠ **Equal Sums 1**

THE CHALLENGE: These three circles create 5 regions. Put 1 to 5 once each in these regions so the sum of the numbers in each circle is the same.

1 2 3 4 5 ♠

2
♦ **Filling Squares With Squares**

Here is how to fill one large square with 1, 4, or 7 squares.

THE CHALLENGE: Find other square counts for filling a large square. Can you do it for 2, 3, 5, 6, 8, 9, or 10 squares?

♦

2
♥ **Lines 2**

These five connected line segments go through all nine points of this 3 x 3 array.

THE CHALLENGE: Find **four** connected line segments that create a path that goes through all nine points.

♥

2
♠ **Finding the Pieces 5**

A **trapezoid** has one pair of parallel sides. The figure below is broken into as few triangles and trapezoids as possible.

THE CHALLENGE: Do the same for these two figures.

♠

3
♠ **Equal Sums 3**

THE CHALLENGE: These three circles create 7 regions. Put 1 to 7 once each in these regions so the sum of the numbers in each circle is the same.

1 2 3 4 5 6 7 ♠

3
♦ **Finding the Pieces 2**

A **trapezoid** has one pair of parallel sides, a **parallelogram** has two pairs.

THE CHALLENGE: Count the number of triangles, trapezoids, and parallelograms in this pyramid.

♦

3
♥ **Square Sums 1**

THE CHALLENGE: Fill in the four missing numbers with numbers from 1 to 7 so the rows and columns add up to the numbers on the outside of the 2 by 2 square.

		3
		12
9	6	+

♥

3
♠ **Stick Areas**

12 sticks are used to enclose a region. These three possibilities enclose areas of size 3, 5, and 9.

THE CHALLENGE: Find all the possible areas you can enclose with 12 sticks. What happens for other numbers of sticks?

♠

4
♣ **Equal Sums 4**

THE CHALLENGE: These four circles create 8 regions. Put 1 to 8 once each in these regions so the sum of the numbers in each circle is the same.

1 2 3 4 5 6 7 8 ♣

4
♦ **Avoiding Rectangles 2**

The X's in the left grid form two rectangles. The X's in the right grid avoid forming any rectangles.

THE CHALLENGE: Place as many X's as you can in this grid avoiding creating any rectangles.

♦

4
♥ **Sum Pyramids 2**

In a **Sum Pyramid**, each number is the sum of the two numbers below it.

THE CHALLENGE: Use the numbers from 1 to 25, at most once, to make a Sum Pyramid.

♥

4
♠ **Parades 2**

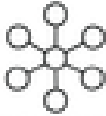
This map of Königsberg shows the island in the middle of the river and the seven bridges that span the river.

THE CHALLENGE: If you can, find a parade route that crosses each of the bridges exactly once. If you can't, give a reason why.

♠

♣ Flowers 1

The sums in a **Magic Flower** are the same for all straight lines of three circles.



THE CHALLENGE: Use the numbers 1 to 7 to make Magic Flowers using this diagram. Is there more than one way to do it?

♣ 9

♦ Triangles

A **regular triangle** is a triangle with equal sides and equal angles.



THE CHALLENGE: Remove the fewest dots so no regular triangles of any size or orientation are formed by the remaining dots.

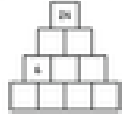
♦ 9

♥ Pyramids 4

In a **Sum Pyramid**, each number is the sum of the two numbers below it.



THE CHALLENGE: Use the numbers from 1 to 24, at most once, to complete this Sum Pyramid.



♥ 9

♠ Ten Numbers



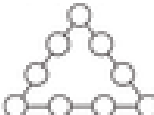
THE CHALLENGE: You have five bags of coins. Each has one kind of coin. The bags have coins worth 1, 3, 5, 7, and 9. If you can, find ten coins that add up to 43. If you can't, describe why it is impossible. Which numbers are possible?

♠ 9

♣ Magic Triangles 2

The circles on each side of a **Magic Triangle** add up to the same thing.

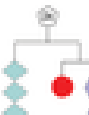
THE CHALLENGE: Use 1 to 9 once each to make Magic Triangles. How many ways can you do it?



♣ 9

♦ Balance Beam 1

The weight is the same on each side of a **balance beam**. The total weight is given in the top shape. The weight for a given shape is always the same.



THE CHALLENGE: Find the weight of each type of shape in this figure.

♦ 9

♥ Difference Pyramids 2



In a **Difference Pyramid**, each number is the difference of the two numbers below it.

THE CHALLENGE: Use the numbers from 1 to 10 once to make a Difference Pyramid.



♥ 9

♠ Fill in the Blanks 2

THE CHALLENGE: Use the numbers from 1 to 8 at most once to fill in these blanks. How many solutions can you find?

$\square + \square = \square + \square = \square - \square$

1 2 3 4 5 6 7 8

♠ 9

♣ Magic Squares 3

In **Magic Squares**, all rows, columns, and diagonals add up to the same number.

THE CHALLENGE: Use the numbers 0 to 8 once each to complete a Magic Square. Is there more than one way?



♣ 9

♦ Letter Substitutions 4

In **Letter Substitution Puzzles**, each letter is a digit from 0 to 9, different letters have different values, and no number has 0 as its leftmost digit.

THE CHALLENGE: Find the value of the letters in these puzzles.

$$\begin{array}{r} A \\ + B B \\ \hline C 4 \end{array} \quad \begin{array}{r} D \\ + E D \\ \hline D E \end{array}$$

♦ 9

♥ Ladybugs Don't Add Up 2

Ladybugs with numbers land on leaves. No two ladybugs on a leaf can add up to another on that leaf. The left leaf is OK; the right has $2 + 4 = 6$.



THE CHALLENGE: Starting at 1, how high can you safely go putting ladybugs on three leaves?

♥ 9

♠ Fill in the Blanks 3

THE CHALLENGE: Use the numbers from 1 to 9 at most once to fill in these blanks. How many solutions can you find?

$\square = \square + \square = \square + \square + \square$

1 2 3 4 5 6 7 8 9

♠ 9

♣ Parentheses 1

Where you put parentheses can change the value of an expression. For example, $5 - 3 + 1$ can become $5 - (3 + 1) = 1$ or $(5 - 3) + 1 = 3$.

THE CHALLENGE: Find places to put parentheses in these two expressions to make each one have the value 6.

$10 - 3 - 5 - 2 + 8 - 4$
 $7 - 4 \times 8 - 3 + 4 \times 2$

♣ 9

♦ Sujiko Puzzle 3

THE CHALLENGE: Fill in this Sujiko Puzzle. Use the numbers from 1 to 9 in the nine squares. Each circled number must be the sum of the four squares surrounding it.



♦ 9

♥ Water Cups 1

You have an unmarked 3-unit and 7-unit cup. Use these cups to create other amounts. For example, create 4 units in the larger cup by filling the 7-unit cup and pouring out 3 units to fill the smaller cup.

THE CHALLENGE: Describe steps for putting 2 units in one of these cups. Can you make other amounts?

♥ 9

♠ Fill in the Blanks 7

THE CHALLENGE: Using the numbers 1 to 9 once each, make this sum as close to 1000 as you can.

$$\begin{array}{r} \square \square \square \\ \square \square \square \\ + \square \square \square \\ \hline \end{array}$$

1 2 3 4 5 6 7 8 9

♠ 9

♣ Last Number Standing

The numbers 1 to 5 are written on a

♣ 9

♦ Product Equals Sum

THE CHALLENGE: If you have five

♦ 9

♥ Pirates with Gold 2

Five smart and greedy pirates want

♥ 9

♠ Fractions 1

THE CHALLENGE: Use 1 to 9 at most

♠ 9

evens. Erase any two numbers to be erased and replaced by their difference. This continues until there is just one number.

THE CHALLENGE: Which values are possible for that last single number? Does your answer change if the numbers go from 1 to 6? What about from 1 to 7?

♣ 6

positive whole numbers minus, when you add them up you get the same answer as when you multiply them, what is the largest possible value of one of those five numbers?

EXPLORATION: What happens for different numbers of numbers in this situation? For example, for two numbers $2 \times 2 = 2 + 2$.

♦ 6

to split up 47 gold coins, even rule is: The youngest pirate proposes a plan. The plan is used if okay with over half of all the pirates. If not, the youngest pirate leaves with no gold, and the new youngest pirate proposes a plan.

THE CHALLENGE: What is the most the youngest pirate can get?

♥ 6

once in these boxes to make the equation true. Is there more than one solution?

$$\frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{\square}$$

1 2 3 4 5 6 7 8 9

♠ 6

10

♣

Broken Calculator 1

This calculator is broken. The only keys that work are: 4, 7, +, and =. It's still possible to make every number. For example $1 = 4 + 4 - 7$.



THE CHALLENGE: Show how to make every number from 1 to 12.

♣ 01

10

♦

Max Product For Sums of 16

THE CHALLENGE: What is the largest product you can make using numbers that add up to 16? Example: $16 = 10 + 6$; $10 \times 6 = 60$ Better: $16 = 6 + 5 + 5$; $6 \times 5 \times 5 = 150$

EXPLORATION: How does your strategy change if you replace 16 with 20, or even 100?

♦ 01

10

♥

Paths on Boards 1

The first board has a path that visits every square starting at the black dot. The second board does not.



THE CHALLENGE: Identify other starting positions that begin paths that can visit every square.

♥ 01

10

♠

Fractions 3

THE CHALLENGE: Use 1 to 9 at most once in these boxes to make this difference as small as possible. How does this change if you allow, or not allow, improper fractions?

$$\square \frac{\square}{\square} - \square \frac{\square}{\square}$$

1 2 3 4 5 6 7 8 9

♠ 01

J

♣

Are Their Hands Tied?

THE CHALLENGE: Two people tie their own hands together loosely with a string. The two strings loop inside each other. Can they get apart without untying their strings?



♣ J

J

♦

Equal Products

THE CHALLENGE: Using the numbers from 1 to 9 at most once, assign seven different numbers to the letters A to G so that these three products are the same.

$$A \times B \times C = C \times D \times E = E \times F \times G$$

♦ J

J

♥

Dominos on Boards

This first board is easy to cover with dominos. The second is impossible.



THE CHALLENGE: Why is one of these impossible to cover?

♥ J

J

♠

Fractions 4

THE CHALLENGE: Use 1 to 9 at most once in these boxes to make the double-digit fraction as close to the Target Number as possible without equalling it. Use Target Numbers starting at 1 and going up to 8.

$$\frac{\square \square}{\square \square} \sim \text{Target}$$

1 2 3 4 5 6 7 8 9

♠ J

Q

♣

Finding Squares 2

THE CHALLENGE: Count the squares of all sizes and orientations in this grid (some will have diagonal sides).



♣ Q

Q

♦

Removing Digits

1234512345123451234512345

THE CHALLENGE: Which ten digits would you remove (they don't need to be next to each other) from this number to make the new number as large as possible? Which ten digits would you remove to make it as small as possible?

♦ Q

Q

♥

Mystery Sums 2

There are 5 boxes to weigh, and each of them weighs under 20 pounds. Unfortunately, the one available scale only weighs things over 20 pounds. The packages, weighed in pairs, weigh 22, 24, 25, 26, 27, 28, 29, 30, 32, and 33 pounds.

THE CHALLENGE: Find out how much each package weighs.

♥ Q

Q

♠

Fractions 7

THE CHALLENGE: Use 1 to 9 at most once in these boxes to make the equation true. How many solutions can you find using proper fractions?

$$\frac{\square}{\square} + \frac{\square}{\square} = \frac{\square}{\square}$$

1 2 3 4 5 6 7 8 9

♠ Q

K

♣

Handshakes at a Party 2

Ten people were at a party. A lot of

K

♦

Moving Digits 2

If you reverse the digits for 2734 you

K

♥

Combining Digits 1 2 3 4

Here is a way to get 0 and 1 using

K

♠

Fractions 12

THE CHALLENGE: Use 1 to 9 at most

handmaking took place, when asked how many hands each person shook, they discovered that each number was different. One person yelled out "That's impossible!"

THE CHALLENGE: Was that person right, did someone make a mistake in their handshake count? How do you know?



get the number 4372.

THE CHALLENGE: Find a 4-digit number that, when you multiply it by 4, you reverse its digits.

EXPLORATION: Why doesn't this happen for numbers less than 1000? Also, look for numbers larger than 1,999 that have this property.



each of 1, 2, 3, and 4 exactly once.

$0 = 1 + 4 - (2 + 3)$
 $1 = (2 - 1) \times (4 - 3)$

THE CHALLENGE: Starting at 0, how many numbers can you get using all the numbers 1, 2, 3, and 4 in any order, using addition, subtraction, and multiplication?



once in these boxes to make the expression 1) equal to $2/3$, and again 2) as close to $5/11$ as possible.

$\frac{\square}{\square} \times \frac{\square}{\square} = \frac{2}{3}$
 $\frac{\square}{\square} \times \frac{\square}{\square} = \frac{5}{11}$

1 2 3 4 5 6 7 8 9



Joker

Parallel lines have a lot in common ...
it's a shame they'll never meet.

How do you keep warm in a cold room? ...
go to the corner - it's always 90 degrees.

Joker

Joker

I saw my math teacher looking suspicious with a piece of graph paper ...
the teacher must be plotting something.

A farmer had 197 cows out in the field ...
but when he rounded them up he had 200!

Joker

Grades 2-5 Math Puzzles

These puzzles are for grades 2 to 5. They can also be enjoyed by "children" of all ages. Get solutions, notes, card face images, and more detailed versions of each puzzle at this link.



www.earlyfamilymath.org/grades-2-5

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